TeraMove-H — Walkthrough

Plain-language guide to the paper, with every formula
(main focus: user association)

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1 The idea

THz access points (APs) need line of sight, and the people they serve are also what blocks them. TeraMove fixed blockage by *moving* APs. TeraMove-H tries two cheaper fixes first:

1. a **Wi-Fi layer** together with smarter **user association**;

2. **power control**: idle APs are switched off, and busy APs use less power.

APs move only if these electronic fixes are not enough.

2 Setup

Building	$32 \times 20 \times 3$ m floor, 12 rooms
THz APs	14 APs from the MILP (Fig. 1); state $\theta_a = (x_a, y_a, z_a, \phi_a, \alpha_a)$
AP height	$z_a \in \{2.0, 2.2, \ldots, 3.0\}$ m
AP azimuth	ϕ_a in 45° steps, compass bearing (0° = North, clockwise)
Beam width	$\alpha_a \in \{30, 45, 60, 75, 90, 120\}^\circ$
AP movement	along a ceiling rail inside its own room only
Wi-Fi APs	4 fixed ceiling APs, 6 GHz, 160 MHz each, separate channels (no mutual interference)
People	80 % in 6 Gaussian clusters ($\sigma = 1$ m), 20 % uniform; bodies never overlap
Body model	cylinder, radius 0.2 m, height 1.6 m
Users	each person is an active user with probability 0.5; device at 1.0 m, random facing direction
Demand	every user needs at least $R_{\min}$ (default 1 Gb/s)

3 Link model

3.1 THz path loss (dB)

Path loss

$$L_{a,u} = \underbrace{20 \log_{10} \frac{4\pi f d}{c}}_{\text{free space}} + \underbrace{4.343\, k(f)\, d}_{\text{absorption}} + \underbrace{L^{\text{st}}_{a,u}}_{\text{walls}} + \underbrace{\min\!\left(n_{a,u} L_h,\ L^{\max}_h\right)}_{\text{bodies}} + \underbrace{L_{\text{sb}}\, \mathbf{1}^{\text{sb}}_{a,u}}_{\text{own body}}$$

Why: Each term is a separate reason the signal gets weaker, so the losses (in dB) simply add up. *Free space*: the wave spreads over a growing sphere, so power falls with d^2; at 300 GHz this is already about 82 dB at 1 m. *Absorption*: water vapour in the air soaks up THz energy, a fixed number of dB per metre. *Walls*: each wall crossed costs a fixed amount that depends on its material. *Bodies*: a person in the path costs about 20 dB; we cap the total at 40 dB because after two bodies the link is dead anyway. *Own body*: if the AP is behind you, your own body blocks it.

Symbol	Meaning	Value
f	carrier frequency	300 GHz
d	3-D AP–user distance	–
c	speed of light	–
$k(f)$	molecular absorption (ITU-R P.676)	$1.54 \times 10^{-3}\,\mathrm{m}^{-1}$ (6.68 dB/km)
L^{st}	wall penalty per wall crossed	concrete 30, wood 20, glass 15 dB
$n_{a,u}$	bodies whose circle the ray crosses at height ≤ 1.6 m	–
$L_h,\ L_h^{\max}$	body loss, cap	20 dB, 40 dB
$\mathbf{1}^{\mathrm{sb}}$	AP lies behind the user's own body (3GPP self-blocking region)	0 or 1
L_{sb}	self-blocking loss	30 dB

Linear path gain:

$$\ell_{a,u} = 10^{-L_{a,u}/10}.$$

Why: Powers from different APs must be *added*, and adding only works on linear values, not on dB.

3.2 AP antenna gain (energy conserving)

The main lobe covers the lower half-space within $\pm\alpha/2$ of ϕ_a. Its solid angle is α steradians, so

AP gain

$$G_m(\alpha) = \frac{4\pi - g_s\,(4\pi - \alpha)}{\alpha}, \qquad g_s = -10 \text{ dBi (side lobe)}, \quad \alpha \text{ in radians.}$$

Why: An antenna cannot create energy; it only redirects it. Averaged over all directions the gain must be 1 (i.e. 4π when integrated), so $G_m \cdot \alpha + g_s(4\pi - \alpha) = 4\pi$. Solving for G_m gives the formula: squeeze the beam (small α) and the main gain goes up.

A wider beam gives less gain: **13.4 / 10.4 / 7.4 dBi** for $\alpha = 30° / 60° / 120°$. The AP gain towards user u is

$$G^t_{a,u} = \begin{cases} G_m(\alpha_a), & \Delta\phi_{a,u} \leq \alpha_a/2, \\ g_s, & \text{otherwise.} \end{cases}$$

Why: A user inside the sector gets the full beam; outside it, only the weak side-lobe leakage.

3.3 User device antenna

The device points a $G_r = 20$ dBi pencil beam at its serving AP. The same energy rule fixes the main-lobe cone, with half-angle ψ:

$$\Omega_r = \frac{4\pi(1 - g_r)}{G_r - g_r}, \qquad \psi = \arccos\!\Big(1 - \frac{\Omega_r}{2\pi}\Big) = 10.9°.$$

Why: Same energy rule for the phone's pencil beam. A cone of half-angle ψ covers $2\pi(1 - \cos\psi)$ steradians, which gives ψ. Only APs inside this narrow cone can interfere strongly with the user.

$G^r_{u,a,b}$ is the device gain towards AP b when the beam points at AP a. It equals G_r if b lies within ψ of the direction to a, and $g_r = -10$ dBi otherwise.

3.4 SINR and capacity

Write $h_{b,u} = G^t_{b,u}\,\ell_{b,u}$ for the channel gain.

SINR, noise and capacity

$$\mathrm{SINR}_{u,a} = \frac{p_a\,h_{a,u}\,G_r}{N_0 + \sum_{b\neq a} p_b\,h_{b,u}\,G^r_{u,a,b}}$$

$$N_0 = -174 + 10\log_{10} B + \mathrm{NF} = -174 + 100 + 10 = -64 \text{ dBm}$$

$$c_{u,a} = \begin{cases} B\,\min\big(\log_2(1+\mathrm{SINR}_{u,a}),\,6\big), & \mathrm{SINR}_{u,a} \geq \gamma_{\mathrm{th}} = 5 \text{ dB}, \\ 0, & \text{otherwise}, \end{cases} \qquad B = 10 \text{ GHz}.$$

Why: *SINR*: useful power divided by everything unwanted (noise plus the power of all other APs at this user). *Noise*: thermal noise is −174 dBm per Hz, so a wide 10 GHz channel collects +100 dB of it, plus 10 dB added by the receiver. *Capacity*: Shannon's formula gives bits/s/Hz $= \log_2(1 + \mathrm{SINR})$; we cap it at 6 (64-QAM, the best practical modulation) and set it to 0 below 5 dB, where a THz link cannot be held.

3.5 Wi-Fi link

3GPP InH-Office path loss, with $f_c = 6$ GHz:

$$\mathrm{PL}_{\mathrm{LOS}} = 32.4 + 17.3\log_{10} d + 20\log_{10} f_c,$$

$$\mathrm{PL}_{\mathrm{NLOS}} = \max\big(\mathrm{PL}_{\mathrm{LOS}},\; 17.3 + 38.3\log_{10} d + 24.9\log_{10} f_c\big).$$

Why: These are standard 3GPP curves fitted to indoor measurements. The distance exponent is 1.73 in line of sight and 3.83 when walls or people are in the way. The max makes sure a blocked link is never better than a clear one.

A link is LOS only if no full-height wall and no body is in the way. The rate is the 2-stream 160 MHz PHY rate of the highest 802.11ax MCS whose sensitivity is met, times a MAC efficiency of 0.7. The maximum Wi-Fi rate is about **1.68 Gb/s**.

4 User association (the core)

4.1 Airtime

Every AP time-shares its channel. If user u is served by AP a, it needs this share of the AP's time:

Airtime needed

$$\tau_{u,a} = \frac{R_{\min}}{c_{u,a}}$$

Why: To deliver $R_{\min}$ over a link of speed c, the AP must spend a fraction $R_{\min}/c$ of its time on that user. Example: 1 Gb/s on a 40 Gb/s link needs 2.5 % of the time.

Each AP can give out at most **1** unit of time in total.

Link	Capacity	Airtime needed at 1 Gb/s
Good THz	40 Gb/s	0.025
Good Wi-Fi	1.5 Gb/s	0.67 (two-thirds of the whole AP)

4.2 Optimisation problem

Let $x_{u,a} = 1$ if user u is served by AP a, and 0 otherwise.

Association problem (maximum GAP)

$$\begin{aligned}
\max_{\mathbf{x}} \quad & \sum_u \sum_a x_{u,a} \\
\text{s.t.} \quad & \sum_a x_{u,a} \le 1 && \forall u \quad \text{(at most one AP per user)} \\
& \sum_u \tau_{u,a}\, x_{u,a} \le 1 && \forall a \quad \text{(airtime budget)} \\
& x_{u,a} = 0 \ \text{ if } c_{u,a} < R_{\min} && \text{(link too weak)} \\
& x_{u,a} \in \{0,1\}
\end{aligned}$$

Why: We count served users. Each user takes one AP at most; an AP cannot hand out more than 100 % of its time; and a link slower than $R_{\min}$ is useless even with all the time.

Satisfaction is the fraction of users who get at least $R_{\min}$:

$$S = \frac{1}{|\mathcal{U}|} \sum_u \sum_a x_{u,a}.$$

Why: Satisfaction is simply the share of users who got what they asked for.

This is a **maximum Generalised Assignment Problem (GAP)** in which every user has the same value. Think of packing items (users) into bins (APs), where an item's size depends on the bin it goes into.

4.3 Why it is hard (Proposition 1)

Deciding whether *all* users can be served is NP-complete. It contains the PARTITION problem: given integers $s_1, \ldots, s_k$ with $\sum_i s_i = 2T$, can they be split into two halves of equal sum? Use two APs and choose the capacities so that

$$c_{u,1} = c_{u,2} = \frac{R_{\min}\, T}{s_u} \quad \Longrightarrow \quad \tau_{u,a} = \frac{s_u}{T}.$$

Both airtime budgets hold with everyone served **if and only if** the integers split into two halves that each sum to T. So an exact solver is too slow for every evaluation, and we use a greedy algorithm.

Why: If an exact solver could always answer quickly, it would also solve PARTITION quickly, which is believed impossible. So we need a fast heuristic.

4.4 Greedy algorithm (`gap_greedy`)

1. List every usable (user, AP) pair, THz and Wi-Fi together.
2. Sort the pairs by airtime $\tau_{u,a}$, smallest first.
3. Walk down the list. Assign a pair only if the user is still free **and** the AP still has enough time left.
4. Users left over are unserved.

The cost is $O(KM \log KM)$ for K users and M APs. Cheapest pairs first leaves room for the most users. With a single AP this rule is exactly optimal.

Why: Short tasks first: giving time to the users who need the least of it leaves the most time for others, like packing small items into a suitcase first. Because THz and Wi-Fi pairs are in one list, a blocked user automatically falls back to Wi-Fi, and a full THz AP pushes users to a second-best AP.

Worked example ($R_{\min} = 1$ Gb/s).

User	THz-A	Wi-Fi
u1	20 Gb/s $\rightarrow \tau = 0.05$	1.6 Gb/s $\rightarrow \tau = 0.63$
u2	4 Gb/s $\rightarrow \tau = 0.25$	1.2 Gb/s $\rightarrow \tau = 0.83$
u3	blocked	1.5 Gb/s $\rightarrow \tau = 0.67$

Sorted pairs: (u1,A) 0.05 $\rightarrow$ (u2,A) 0.25 $\rightarrow$ (u1,W) 0.63 $\rightarrow$ (u3,W) 0.67 $\rightarrow$ (u2,W) 0.83.

Step	Pair	Action	Load A	Load W
1	(u1, A)	assign	0.05	0
2	(u2, A)	assign	0.30	0
3	(u1, W)	skip (u1 already served)	0.30	0
4	(u3, W)	assign	0.30	0.67
5	(u2, W)	skip (u2 already served)	0.30	0.67

All 3 users are served, and Wi-Fi is spent only on the blocked user.

4.5 Baselines

Rule	How it decides	Weakness
Max-SINR (old TeraMove)	strongest THz AP; no Wi-Fi	blocked users are lost; ignores load
Threshold offload	strongest THz AP if SINR $\geq$ 5 dB, else best Wi-Fi	ignores load; cannot pick a second-best THz AP
Greedy GAP (TeraMove-H)	cheapest airtime first, over all APs	not always optimal (see below)

4.6 How good is the greedy?

It was compared with the exact integer program, solved by HiGHS (`scipy.optimize.milp`), on **450** instances:

Check	Result
Greedy equals the optimum	**90.9 %** of instances
Worst case	92.4 % of the optimum (5 users short)
Median time: greedy vs exact	**0.29 ms** vs 32.9 ms

4.7 When association matters

Gain of the greedy GAP over threshold offload, in satisfaction:

$R_{\min}$ (Gb/s)	0.1	0.25	0.5	1	2
GAP − offload (pp)	+1.5	+1.9	+1.3	+0.8	0.0

- **Low demand.** Each Wi-Fi AP can carry many users, so sharing its time wisely pays off. At 0.1 Gb/s, re-association alone serves **93.2 %** of users, and **78 %** of crowd snapshots need no AP movement at all.
- **High demand.** One Wi-Fi AP tops out at 1.68 Gb/s, so it carries about one user at 1 Gb/s. There is little left to allocate, and the rule barely matters.

5 Power control

5.1 Sleep

Any THz AP with no users is switched off ($p_a = 0$), and users are re-associated. This repeats for up to 3 rounds. A round is kept only if S does not drop.

5.2 Turn spare airtime into lower power

AP a's airtime load is

$$L_a = \sum_{u \in \mathcal{U}_a} \tau_{u,a} \leq 1.$$

Why: L_a is how much of AP a's time is actually used. If $L_a < 1$, the AP has spare time.

If every user's time is stretched by $1/L_a$, the user still gets $R_{\min}$ with a lower capacity:

Target SINR

$$c_u^\star = c_{u,a}\, L_a \qquad \Longrightarrow \qquad \gamma_u = \max\!\left(\gamma_{\text{th}},\; 2^{\,c_u^\star/B} - 1\right)$$

Why: Spare time can be traded for power. Give each user $1/L_a$ times more time, and it can live with a slower link $c_u^\star = c_{u,a}L_a$ and still get $R_{\min}$. Inverting Shannon ($c = B\log_2(1+\gamma)$) gives the SINR needed for that speed; never below 5 dB so the link stays up.

5.3 Yates minimum-power iteration

Power update

$$I_a(\mathbf{p}) = \max_{u \in \mathcal{U}_a} \frac{\gamma_u \left(N_0 + \sum_{b \neq a} p_b\, h_{b,u}\, G^r_{u,a,b}\right)}{G_r\, h_{a,u}}$$

$$\mathbf{p} \;\leftarrow\; \min\!\left(\mathbf{I}(\mathbf{p}),\; P_{\max}\right), \qquad P_{\max} = 20 \text{ dBm}$$

Why: Each AP picks the smallest power that lets its *weakest* user reach its target, given the interference it sees now. Changing one AP's power changes interference for the others, so we repeat until nothing changes. The min with $P_{\max}$ keeps powers legal.

Proposition 2. $\mathbf{I}$ is a *standard interference function*:

Property	Holds because
Positive	$N_0 > 0$
Monotone	all coupling terms are ≥ 0
Scalable: $\mu\,\mathbf{I}(\mathbf{p}) > \mathbf{I}(\mu\mathbf{p})$ for $\mu > 1$	the N_0 term does not scale with μ

At the current powers $\mathbf{p}^0$, every served user already meets its target, so $\mathbf{I}(\mathbf{p}^0) \leq \mathbf{p}^0$. Starting from there, the iteration only goes **down** and converges to the minimum power vector $\mathbf{p}^\star$. At $\mathbf{p}^\star$, every served user keeps $R_{\min}$ and no AP exceeds its airtime budget:

$$c_u \geq c_u^\star \implies \frac{R_{\min}}{c_u} \leq \frac{\tau_{u,a}}{L_a} \implies \sum_u \frac{R_{\min}}{c_u} \leq 1.$$

Why: Every user's new speed is at least the planned $c_u^\star$, so its time need is at most the planned (stretched) share, and these shares add up to at most 1. Hence nobody loses service.

Result: THz power drops from **1.40 W to 0.53 W (−62.1 %)**, with about 4 of the 14 APs switched off. Satisfaction is unchanged.

6 Escalation and search

The stages run in order and stop at the first one that reaches $S \geq \eta = 0.9$:

Stage	What it does	Physical cost
T0	greedy GAP over THz + Wi-Fi at full power	none
T1	sleep + re-associate + minimum power	none
T2	genetic algorithm (GA) over azimuth ϕ and width α (1 budget unit)	none
M	GA over height, position within the room and beams (3 units), then Hungarian matching per room	movement

One budget unit is $40 \times 40 = 1600$ fitness evaluations. Final power control always runs at the end.

6.1 Fitness used by the search

Fitness

$$F = 0.6\,S + 0.4\,Q + 0.5 \cdot \mathbf{1}[S \geq \eta] - \beta M - \gamma C_p$$

Why: S is the main goal. Q gives partial credit, so the search can tell *almost* served from hopeless and climb towards better layouts. The bonus rewards crossing the 90 % target. M punishes moving hardware, and C_p stops APs from bunching up.

$$Q = \frac{1}{|\mathcal{U}|} \sum_u \min\Big(1,\ \frac{\max_a c_{u,a}}{R_{\min}}\Big) \qquad \text{(smooth link quality)}$$

$$M = \frac{\sum_a \|(x_a, y_a, z_a) - (x_a^0, y_a^0, z_a^0)\|}{|\mathcal{T}| \cdot \text{building diagonal}}$$

Why: Total distance moved, divided by its largest possible value, so the penalty stays between 0 and 1.

$$C_p = \frac{1}{\binom{n}{2}} \sum_{i<j} \max\Big(0,\ 1 - \frac{d_{ij}}{d_{\min}}\Big), \qquad d_{\min} = 3 \text{ m},$$

Why: Every pair of APs closer than 3 m adds a penalty that grows as they get closer; averaged over all pairs.

with $\beta = 0.25$ and $\gamma = 0.3$ (TeraMove configuration D).

6.2 Hungarian matching

New AP states are matched to physical APs inside each room so that total movement is minimal:

$$\sigma^\star = \arg\min_\sigma \sum_i C_{i,\sigma(i)}, \qquad C_{ij} = \|\mathbf{r}_i - \mathbf{r}'_j\|.$$

Why: All APs are identical, so it does not matter *which* AP goes to a new spot. The Hungarian algorithm picks the pairing with the least total travel, in $O(n^3)$ time.

6.3 Commissioning

The MILP layout's beams and heights were re-tuned once for the empty floor; the positions did not change. Empty-floor coverage rose from **66.2 % to 100 %**.

7 Main results

30 seeds, $R_{\min} = 1\,\text{Gb/s}$, 120 people.

Method	Users satisfied S	AP movement
Static (no adaptation)	44.8 %	0 m
Old TeraMove	50.9 %	11.8 m
Re-association only (T0)	51.6 %	0 m
T0–T2 (no movement)	58.9 %	0 m
THz only (no Wi-Fi)	77.2 %	25.8 m
Threshold offload	83.0 %	23.3 m
TeraMove-H	**84.4 %**	**24.6 m**
Flat GA (all dimensions at once)	87.2 %	30.3 m

The flat GA satisfies 2.8 pp more users but moves APs about 23 % more. That is a trade-off, not a win for either method.

8 Where it lives in the code

File	What is in it
`teramove_h/association.py`	`gap_greedy`, `gap_exact`, `max_sinr_pref`, `naive_offload_pref`
`teramove_h/evaluator.py`	association → sleep → power control pipeline
`teramove_h/power.py`	SINR targets γ_u and the Yates iteration
`teramove_h/network.py`	SINR and capacity for every (user, AP) pair
`teramove_h/antenna.py`	energy-conserving gains $G_m(\alpha)$ and ψ
`teramove_h/channel.py`	ray geometry, walls, bodies, self-blocking, Wi-Fi rates
`teramove_h/cascade.py`	escalation stages T0 → T1 → T2 → M, and every baseline
`paper/main.tex`	Sec. III (model), Sec. IV-B (association), Sec. V (results)